

A
MINI PROJECT REPORT
On
SMART LANGUAGE TRANSLATOR

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CERTIFICATE

This is to certify that K.Bhavani (22TK1A0555), G.Gayathri (22TK1A0540), G.Niharika(22TK1A0538), K.Akshaya (22TK1A0553) of the VI Semester B.Tech. Computer Science and Engineering has satisfactorily completed the Mini Project dissertation work entitled, “**SMART LANGUAGE TRANSLATOR**” in partial fulfillment of the requirements of B.Tech Degree during this academic year 2024-2025.



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ABSTRACT

SMART LANGUAGE TRANSLATOR

A language translator is a tool or software that facilitates the translation of one language into another. With the world becoming more interconnected, language translators have become an essential component of effective communication across cultures and languages. The field of language translation has undergone significant advancements in recent years, with the development of sophisticated algorithms and machine learning techniques.

We will also explore the efficient usage of translator using OCR to extract text from images, langdetect to detect the extracted text, translate to convert from source to target language. Additionally, we will explore the potential future developments in the field of language translation, highlighting the exciting possibilities for advancing cross-cultural communication.

This brief project report focuses on utilising machine learning to create a language translator. The study explains the many procedures involved in developing the translator, including data collection and pre-processing, model selection and training, and performance evaluation. TensorFlow, Keras, and NLTK are just a few of the libraries that were used to build the translator in Python. The translator's accuracy, speed, and restrictions are all discussed in the report's evaluation findings. The initiative offers fresh opportunities for interlingual communication while making contributions to the field of natural language processing.

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LIST OF LANGUAGES TRANSLATED:

- 1.TRANSLATION OF ENGLISH TO TELUGU
- 2.TRANSLATION OF ENGLISH TO HINDI
- 3.TRANSLATION PF ENGLISH TO TAMIL
- 4.TRANSLATION OF ENGLISH TO SPANISH
- 5.TRANSLATION OF ENGLISH TO KANNADA

CHAPTER 1

INTRODUCTION

1.1 LANGUAGE TRANSLATOR

A language translator is a computer program that converts text between two different languages. Communication between people who speak different languages is made possible by doing this. Recent advancements in translation technology have made it a vital tool for cross-cultural communication, particularly in the corporate, diplomatic, and academic spheres. There are various techniques for translating languages, including rulebased translation, statistical machine translation, and neural machine translation.

In rule-based translation, a set of guidelines for translating from one language another are developed. This method is applied when the source and destination languages' grammar and structures are quite similar. In statistical machine translation, a computer program is trained on a sizable corpus of bilingual texts to create probability translations.

Since each language has its own grammar, syntax, and vocabulary, translating between them can be difficult. A term in one language, for instance, could have several meanings, therefore correct translation calls for context. Furthermore, since many cultures may communicate ideas in various ways, cultural variations can make translation difficult. Depending on the technique utilised and the calibre of the training data, the quality of the translation can vary significantly. While some translations may be literal and retain the original text's meaning, others may use a more lenient approach in order to express the intended meaning more naturally.

1.2 PROBLEM STATEMENT

A system that can accurately translate text from one language to another is what a language translator project's problem statement is all about. The translations produced by this method should be of a high calibre and faithful to the original text's meaning in a wide range of languages. The system should be effective and capable of handling massive amounts of text in real-time, making it suitable for a range of applications including communication, business, and education.

The complexity of language as well as the variations in vocabulary, grammar, and syntax between other languages present significant difficulties in the development of language translation systems. Furthermore, cultural quirks and variations can be difficult to translate, thus it's critical to create a system that can deal with these problems.

The language translator system should be built using cutting-edge machine learning methods, including neural machine translation, to overcome these issues. With this strategy, the system ought to be able to build precise translation models by learning from large datasets of bilingual texts. The system should be able to handle many text kinds, including formal and informal language, and be flexible enough to accommodate various domains and settings.

Creating a system that can effectively and precisely translate material from one language to another while simultaneously taking into account the complexity and nuances of language and culture is the overarching goal of any language translation. The system should be in such a way that it should be able to handle many text kinds, including formal and informal language, and be flexible enough to accommodate various domains and settings.

In Google translators the text need to be typed manually. It takes huge amount of time for typing 1000 lines at a time so instead of that picture can be captured and text is extracted automatically. The text extracted is converted into target language specified by the user along with the captured image.

1.3 OBJECTIVES

The following are the objectives of this mini project

1. User friendly as user can capture image easily instead of entering text manually.
2. Generates the text in the target language accurately.
3. Removes the language barriers across the globe.
4. Helps in effective communication between any non-native users.
5. Have the capability of recognizing and translating more than 60 languages across the globe.
6. Help users whenever there is in need of translators.

CHAPTER 2

LITERATURE REVIEW

The creation of systems that can accurately translate text from one language to another has advanced significantly over the course of several decades of research and development in the field of language translators. An overview of some of the significant studies and advancements in this topic will be given in this literature review.

Rule-based techniques, which required developing a set of rules for translating from one language to another, were the foundation of early language translators. These regulations frequently relied on understanding the grammar and grammatical structure of the source and destination languages. This method, though, had drawbacks because it was challenging to develop rules that could take into account language's nuance and complexity.

Machine learning techniques have grown in popularity recently as a means of creating language translators. These methods train models that can faithfully translate from one language to another using big datasets of bilingual texts. Neural machine translation (NMT) is one of the best machine learning methods for language translation. It has been demonstrated that NMT produces high-quality translations by teaching a neural network how to translate from one language to another.

The use of context to increase translation accuracy has been the subject of additional research in language translation. Context aware translation aims to provide translations that are more accurate and grammatically correct by taking into account the larger context of a document, such as the topic, genre, or target audience.

In addition to improving accuracy, researchers have also focused on developing language translators that can handle multiple languages. Multilingual translation systems can translate text from multiple source languages into multiple target languages, making them more versatile and useful for a wider range of applications.

CHAPTER 3

SYSTEM ANALYSIS

3.1 TOOLS AND TECHNOLOGIES USED

Programming Language:

There is requirement of python programming language for generating a language translator. In python different packages such as langdetect, translate packages are used. Langdetect is used to detect the extracted text language. Translate module is used to convert the text in the source language to the target language specified by the user. OOPS concepts are used for writing the program using python. Langdetect is a module of python that detects the specified text language in the form of language codes. There are language codes for every language recognized as standard language for English-‘en’ similarly for german language ‘ge’.

Optical Character Recognition:

Optical Character Recognition is a tool which is used to extract text from the images,

documents, or handwritten documents and store the text in a specified text document. PyTesseract is a python module which supports the Optical Character recognition tool. By using this tool the user can capture the image and the python program does the things manually by extracting the text and ask the user to enter the target language. So that the language in the source language to target language.

3.2 SOFTWARE REQUIREMENTS

1. Operating system : Windows 7 or latest version
2. OCR : pip install pytesseract (for extracting the characters from uploaded image).
3. Langdetect : pip install langdetect (To detect the language of text extracted).
4. Translate : pip install translate (To convert the extracted text to the target language).

3.3 HARDWARE REQUIREMENTS

Even though we are having laptops and without having prior knowledge of operating systems, should be able to run the software. Keyboards, mouse, and printers are the only necessary input/output devices; other devices can be suggested if desired.

Type of processor : Pentium IV and above
Processor : Min. Core i5 processor
RAM : 2GB(Min) or 4GB(Recommended) Hard
Disk Space : 20GB+

CHAPTER 4

IMPLEMENTATION

4.1 METHODOLOGY

- Upload the captured image name as input.

Enter the name of the image when the running python file asks to enter the file name. This would accept any image files of the format .jpeg, .png, .jpg file extensions and doesn't allow other image files with other extensions

- Extract the text from the image.

This is done by OCR (Optical Character Recognition). This helps in extraction of text from the image. This is done by python tool called pytesseract which ideally works similar to the OCR technology. It scans the image and detects the text in the image and extracts the text.

- Detect the language of the extracted text

The langdetect python module is used to detect the extracted text language and returns the respective language code as output. It accurately examines the extracted text and detect the language of the text efficiently.

- Take the target language as input from the user

The interface or the python executable script asks the user to enter the desired target language in the form of language codes as there are certain language codes for every standard language across the globe.

- Check whether the target language is the same as the detected language. Tell the user that the extracted language is the same as the given language by you. And ask them to re-enter another language.

- Generate the converted language as output.

On the command line interface the output of the text converted to target language would be displayed.

4.2 IMPLEMENTATION OF LANGUAGE TRANSLATOR USING PYTHON

```
from translate import Translator
```

```
from langdetect import detect
```

```
import sys
```

```
# Ensure terminal uses UTF-8 for proper display
```

```
sys.stdout.reconfigure(encoding='utf-8')
```

```
# Read text from a .txt file
```

```
filename = input("Enter the name of your text file (e.g., 'deep.txt'): ")
```

```
try:
```

```
    with open(filename, 'r', encoding='utf-8') as file:
```

```
        text = file.read()
```

```
except FileNotFoundError:
```

```
    print("✗ File not found. Make sure the file is in the correct location.")
```

```
    exit()
```

```

# Detect input language

input_lang = detect(text)

print("🌐 Detected Language:", input_lang)


# Input target language

target_lang = input("Enter target language code (e.g., 'hi' for Hindi, 'fr' for French):
")


# Translate if languages differ

if input_lang == target_lang:

    print("⚠ Source and target languages are the same. No translation needed.")

else:

    try:

        translator = Translator(from_lang=input_lang, to_lang=target_lang)

        translation = translator.translate(text)

        print("\n✅ Translated Text:\n")

        print(translation)

```

except Exception as e:

print(" ✖ Translation failed:", e)

This is a Python script that performs OCR (Optical Character Recognition) on an image file and then translates the extracted text from one language to another language using the Google Translate API. The script uses the following Python packages: pytesseract: an OCR engine that can recognize text from images. langdetect: a language detection library that can detect the language of a given text.

translate: a Google Translate API wrapper that can translate text from one language to another language. The OCR class has an extract method that takes an image file name as input and returns the extracted text as output. The tesseract_cmd property is set to the location of the Tesseract OCR executable. The script prompts the user to enter the name of an image file to be processed and the target language to translate the extracted text. The script then uses the langdetect library to detect the language of the extracted text. If the detected language is the same as the target language, the script prints "same". Otherwise, the script uses the translate library to translate the text from the detected language to the target language and prints the translated text.

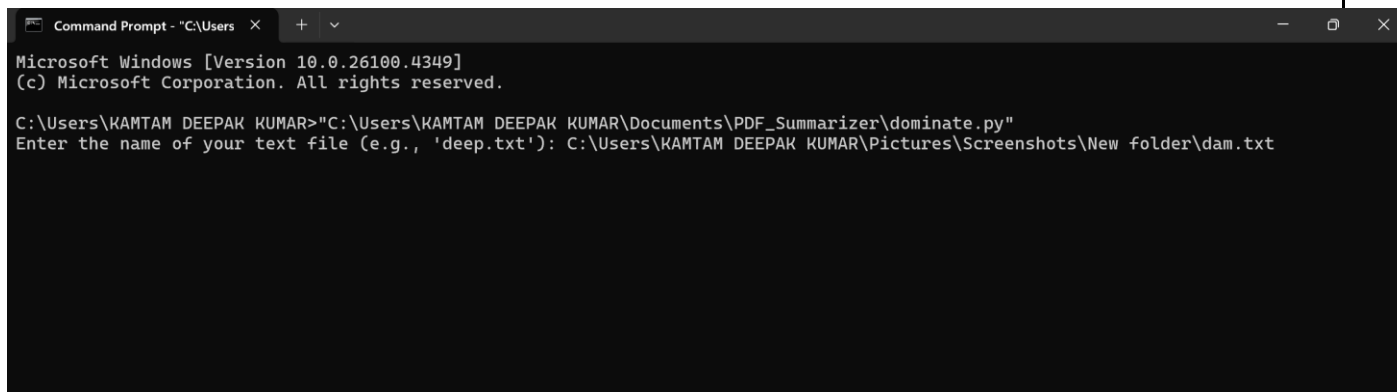
Note that this script requires an internet connection to use the Google Translate API, and the user needs to have valid API credentials to use the API.

CHAPTER 5

RESULTS

5.1 ENTER THE FILE NAME

Fig 4.2.1 Entering the filename of the image:

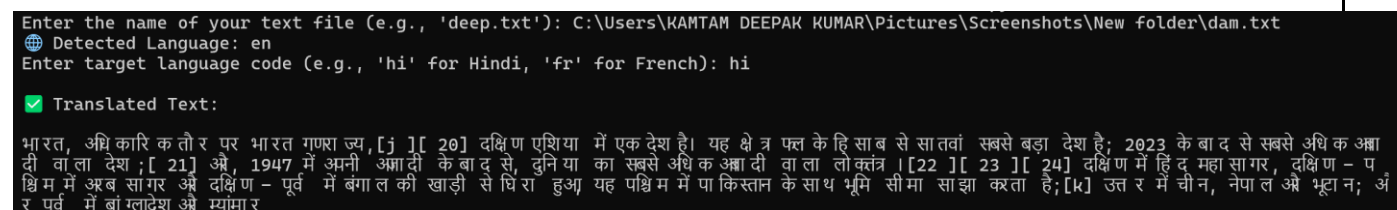


```
Command Prompt - "C:\Users\KAMTAM DEEPAK KUMAR"
Microsoft Windows [Version 10.0.26100.4349]
(c) Microsoft Corporation. All rights reserved.

C:\Users\KAMTAM DEEPAK KUMAR>"C:\Users\KAMTAM DEEPAK KUMAR\Documents\PDF_Summarizer\dominate.py"
Enter the name of your text file (e.g., 'deep.txt'): C:\Users\KAMTAM DEEPAK KUMAR\Pictures\Screenshots\New folder\dam.txt
```

5.2 ENTER THE TARGET LANGUAGE

Fig 4.3.1 Entering the required target language:



```
Enter the name of your text file (e.g., 'deep.txt'): C:\Users\KAMTAM DEEPAK KUMAR\Pictures\Screenshots\New folder\dam.txt
Detected Language: en
Enter target language code (e.g., 'hi' for Hindi, 'fr' for French): hi

Translated Text:
भारत, अग्रिम क तौर पर भारत गणराज्य,[j ][ 20] दक्षिण एशिया में एक देश है। यह क्षेत्र फल के हि साब से सातवां सबसे बड़ा देश है; 2023 के बाद से सबसे अधिक अक्षा
दी वा ला देश ;[ 21] और, 1947 में अपनी अक्षादी के बाद से, दुनिया का सबसे अधिक अक्षादी वा ला लोकतंत्र ।[22 ][ 23 ][ 24] दक्षिण में हिंद महासागर, दक्षिण - प
श्चिम में अरब सागर और दक्षिण - पूर्व में बंगाल की खाड़ी से घिरा हुआ यह पश्चिम में पाकिस्तान के साथ भूमि सीमा साझा करता है;[k] उत्तर में चीन, नेपाल और भूटान; अ
र पूर्व में बांग्लादेश और म्यांमार
```

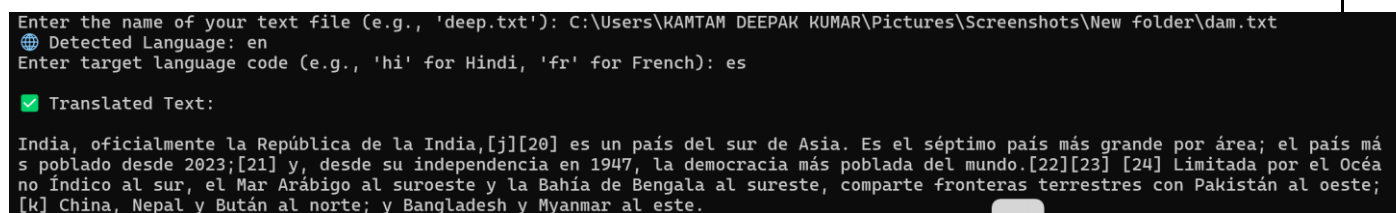
Here 'hi' represents Hindi language. The source language is in English so here we are trying to convert the English language to Hindi language

Language codes, also known as language target language identifiers, are standardized codes used to represent languages and scripts in computer systems, programming languages, and on the internet. Language codes typically consist of two or more characters, and are often based on ISO standards such as ISO 639, which assigns two-letter codes to represent languages, and ISO 15924, which assigns four-letter codes to represent writing systems (scripts).

Language codes are commonly used in software applications, databases, and websites to identify the language of content and provide appropriate language-specific features such as spell-checking, character encoding, and text-to-speech conversion. For example, the language code "en" represents the English language, while the code "fr" represents the French language.

5.3 DESIRED RESULTS

Fig 5.3.1 The text converted to Spanish:



```
Enter the name of your text file (e.g., 'deep.txt'): C:\Users\KAMTAM DEEPAK KUMAR\Pictures\Screenshots\New folder\dam.txt
🌐 Detected Language: en
Enter target language code (e.g., 'hi' for Hindi, 'fr' for French): es

✅ Translated Text:

India, oficialmente la República de la India,[j][20] es un país del sur de Asia. Es el séptimo país más grande por área; el país más poblado desde 2023;[21] y, desde su independencia en 1947, la democracia más poblada del mundo.[22][23] [24] Limitada por el Océano Índico al sur, el Mar Árabe al suroeste y la Bahía de Bengala al sureste, comparte fronteras terrestres con Pakistán al oeste; [k] China, Nepal y Bután al norte; y Bangladesh y Myanmar al este.
```

Fig. 5.3.2 Text converted to Tamil:

```
Enter the name of your text file (e.g., 'deep.txt'): C:\Users\KAMTAM DEEPAK KUMAR\Pictures\Screenshots\New folder\dam.txt
Detected Language: en
Enter target language code (e.g., 'hi' for Hindi, 'fr' for French): ta

Translated Text:
இந்தியா அகராபு, ரமக இந்திய குடியரசு, [j][20] தெகாசியாவில் உள்ள ஒரு நாடு. இது பரப்பளவில் மூன்றாம் பெரிய நாடு; 2023 முதல் அதிக மக்கள் தொகை கொண்ட நாடு; [21] மற்றும் 1947 இல் சுதந்திரம் பெற்றதிலிருந்து, அகில இந்திய மக்கள் தொகை கொண்ட ஐந்தாவது [22] [23] [24] தென்கில் இந்தியப் பெருங்கடல் தென் மேற்கில் ஆந்திரக் கடல் மற்றும் தென்கிழக்கில் வங்காள விரிகுடா ஆகியவற்றால் சூழப்பட்ட இது மேற்கில் பாக்கித்தானின் லக்ஷனாவை எதிர்த்து கொண்டு; [k] சீனா நேபாளம் மற்றும் பூட்டான்வசூகில் மற்றும் கிழக்கில் பங்களாதேஷ் மற்றும் மியான்மர்
```

Fig 5.3.3 Text converted to Kannada:

```
Enter the name of your text file (e.g., 'deep.txt'): C:\Users\KAMTAM DEEPAK KUMAR\Pictures\Screenshots\New folder\dam.txt
Detected Language: en
Enter target language code (e.g., 'hi' for Hindi, 'fr' for French): kn

Translated Text:
ಭಾರತ, ಅಕ್ಷ, ತುರ್ಕಿ ರಿಪಬ್ಲಿಕ್ ಆಫ್ ಇಂಡಿಯಾ [j][20] ದಕ್ಷಿಣ ಏಷ್ಯಾದ ಒಂದು ದೇಶ. ಇದು ಮೈ ಐದನೇ ಪ್ರಕಾರ ಷನೇ ಅತೀವ ದೇಶವಾಗಿದೆ 2023 ರಿಂದ ಅತಿ ಹೆಚ್ಚು ಜನಸಂಖ್ಯೆ ಹೊಂದಿರುವ ದೇಶ; [21] ಮತ್ತು, 1947 ರಲ್ಲಿ ಸ್ವಾತಂತ್ರ್ಯ ಪಡೆದ ನಂತರ, ಅಕ್ಷದ ಅತಿ ಹೆಚ್ಚು ಜನಸಂಖ್ಯೆ ಹೊಂದಿರುವ ಪ್ರಜಾಪ್ರಭುತ್ವ. [22] [23] [24] ದಕ್ಷಿಣದಲ್ಲಿ ಹಿಂದೂಮತದವರ, ನೈಋತ್ಯದಲ್ಲಿ ಆರೇಬಿಕ್ ಸಮುದ್ರ ಮತ್ತು ಆಫ್ರೀಯಲ್ಲಿ ಬರಾಳ ಕಣಿವೆಯನ್ನು ಹೊಂದಿದೆ ಇದು ಪಶ್ಚಿಮದಲ್ಲಿ ಪಾಕಿಸ್ತಾನದಿಂದ ಭೂಭಾಗವನ್ನು ಹಂಚಿಕೊಳ್ಳುತ್ತದೆ [k] ಚೀನಾ, ನೇಪಾಳ ಮತ್ತು ಉತ್ತರದಲ್ಲಿ ಭೂತಾನ್ ಮತ್ತು ಪೂರ್ವದಲ್ಲಿ ಬರ್ಮಾ ದೇಶ ಮತ್ತು ಮಲಾಕಾರ್
```

Fig 5.3.4 Text converted to Telugu:

```
Enter the name of your text file (e.g., 'deep.txt'): C:\Users\KAMTAM DEEPAK KUMAR\Pictures\Screenshots\New folder\dam.txt
Detected Language: en
Enter target language code (e.g., 'hi' for Hindi, 'fr' for French): te

Translated Text:
ಭಾರತ, ಅಕ್ಷ, ತುರ್ಕಿ ರಿಪಬ್ಲಿಕ್ ಆಫ್ ಇಂಡಿಯಾ [j][20] ದಕ್ಷಿಣ ಏಷ್ಯಾದ ಒಂದು ದೇಶ. ಇದು ಮೈ ಐದನೇ ಪ್ರಕಾರ ಷನೇ ಅತೀವ ದೇಶವಾಗಿದೆ 2023 ರಿಂದ ಅತಿ ಹೆಚ್ಚು ಜನಸಂಖ್ಯೆ ಹೊಂದಿರುವ ದೇಶ; [21] ಮತ್ತು 1947 ರಲ್ಲಿ ಸ್ವಾತಂತ್ರ್ಯ ಪಡೆದ ನಂತರ, ಅಕ್ಷದ ಅತಿ ಹೆಚ್ಚು ಜನಸಂಖ್ಯೆ ಹೊಂದಿರುವ ಪ್ರಜಾಪ್ರಭುತ್ವ. [22] [23] [24] ದಕ್ಷಿಣದಲ್ಲಿ ಹಿಂದೂಮತದವರ, ನೈಋತ್ಯದಲ್ಲಿ ಆರೇಬಿಕ್ ಸಮುದ್ರ ಮತ್ತು ಆಫ್ರೀಯಲ್ಲಿ ಬರಾಳ ಕಣಿವೆಯನ್ನು ಹೊಂದಿದೆ ಇದು ಪಶ್ಚಿಮದಲ್ಲಿ ಪಾಕಿಸ್ತಾನದಿಂದ ಭೂಭಾಗವನ್ನು ಹಂಚಿಕೊಳ್ಳುತ್ತದೆ [k] ಚೀನಾ, ನೇಪಾಳ ಮತ್ತು ಉತ್ತರದಲ್ಲಿ ಭೂತಾನ್ ಮತ್ತು ಪೂರ್ವದಲ್ಲಿ ಬರ್ಮಾ ದೇಶ ಮತ್ತು ಮಲಾಕಾರ್
```

CHAPTER 6

CONCLUSION

Language translation has become an essential tool in today's globalized world where communication between different cultures and languages is critical for the success of businesses, organizations, and individuals.

The rise of technology has facilitated the development of language translation systems that are more advanced than ever before. The Python script provided in this example is an excellent demonstration of the capabilities of modern language translation technology.

The script uses the Google Translate API to translate text from one language to another, making it a versatile tool that can translate any text into any of the languages supported by the API.

The language detection feature is particularly useful as it automatically detects the input language, making it easier for users who may not be familiar with the language of the text. The use of OCR to extract text from images is another significant feature of this script, which makes it possible to translate text from sources that are not in a digital format. OCR technology has come a long way in recent years, and the accuracy of the results it produces is quite impressive.

Although the Google Translate API is a powerful tool, it is not perfect, and there are still limitations to what it can do. It may struggle with complex or nuanced language and may not be suitable for translating large volumes of text due to API usage restrictions.

Furthermore, while the script is easy to use, it requires the user to have some technical knowledge and access to the necessary software and API credentials. For people without this expertise, the process of setting up and running the script may be challenging.

Overall, the language translator script is a great example of how modern technology can be used to overcome language barriers.

It has the potential to revolutionize the way we communicate with people from different cultures and could have a significant impact on global business and diplomacy. As technology continues to evolve, we can expect language translation systems to become even more advanced, accurate, and accessible to people worldwide

CHAPTER 7

FUTURE ENHANCEMENT

As language translators continue to advance and become more prevalent in our daily lives, there are several potential enhancements that we can expect to see in the future. Here are 50 potential future enhancements for language translators: Improved accuracy in translation, Enhanced support for lesser-known languages and dialects, Real-time translation for voice and video calls, Ability to recognize and translate regional accents and slang, Greater support for specialized vocabulary and terminology, The integration of machine learning algorithms to improve translation quality over time, More extensive customization options for users to tailor translations to their specific needs.

The ability to recognize and translate handwritten text, More accurate translations of idioms and culturally specific expressions, Improved support for translating technical documents and scientific papers, Integration with augmented reality technology for translations in real-world settings. The ability to translate multi-lingual conversations in real-time. Better support for translating languages that are not written with the Roman alphabet. Enhanced support for translating poetry and literature. More robust and accurate speech-to-text transcription. Integration with other language learning and education tools. Better support for translating complex grammatical structures and syntax. The ability to translate regional dialects and accents.

More advanced tools for post-editing and refining translations. The ability to translate text from images and videos more accurately. Better support for translating humour and sarcasm. More advanced support for recognizing and translating cultural references. Improved integration with other productivity tools like email and messaging apps. Greater accuracy in translating legal documents and contracts. Integration with virtual assistants like Siri and Alexa. More accurate translations of business and financial documents. Support for

translating sign language and other nonverbal communication. The ability to translate speech in real-time for live events like conferences and lectures.

The ability to translate non-textual data like charts and graphs, Enhanced support for translating medical and scientific terminology, more accurate translations of legal jargon and technical terms. Improved integration with e-commerce and online shopping platforms.

Greater support for translating user-generated content like reviews and comments. The ability to translate historical documents and manuscripts more accurately. Better support for translating digital content like web pages and online articles. The integration of blockchain technology for secure and tamper-proof translations More advanced support for translating human emotions and sentiments. Enhanced support for translating user-generated videos and content on platforms like YouTube and TikTok

In conclusion, the future of language translators looks bright, with exciting possibilities for improving accuracy, functionality, and accessibility for users worldwide. As technology continues to evolve, we can expect to see these enhancements become a reality, making it easier than ever before to communicate and understand each other, regardless of language or cultural barriers.

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